22/07/2026

Code-B_Phase-2

Aakash

Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

Fortnightly Test for NEET-2027 (XII Passed) Ph-2 - FT02B

Time : 180 Min.

Topics Covered:
Physics : Motion in a Plane, Laws of Motion
Chemistry : Structure of Atom, Classification of Elements and Periodicity in Properties
Botany : Cell Cycle & Cell Division, The Living World
Zoology : Biomolecules-II: (Proteins, types & functions, Lipids, Nucleic acids, Enzymes, Cofactors), Breathing & Exchange of Gases-I: (Upto mechanism of breathing)

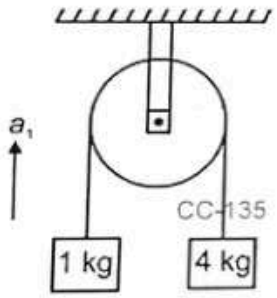
General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

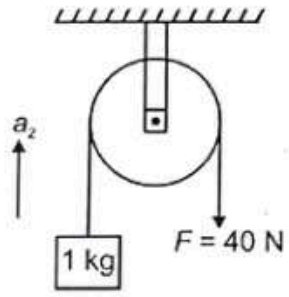
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PHYSICS

1. The position vector of a particle $(\vec{r})$ as a function of time t (in s) is given as $\vec{r} = \left(2t\,\hat{i} + 2t^2\hat{j}\right)$m. The initial acceleration of the particle will be
   (1) $(2\hat{j})\text{m/s}^2$ (2) $(3\hat{j})\text{m/s}^2$
   (3) $(\hat{j})\text{m/s}^2$ (4) $(4\hat{j})\text{m/s}^2$

2. The ratio $\frac{a_1}{a_2}$ of accelerations of 1 kg masses in the two systems shown in the figures is (Pulleys and strings are ideal and g = 10 m/s^2)





   (1) $\frac{1}{5}$ (2) $\frac{1}{3}$
   (3) $\frac{3}{5}$ (4) $\frac{1}{3}$

3. The equation of trajectory of a projectile is given by $y = x - 10x^2$. The maximum height of the projectile is (x and y are in metre)
   (1) 2.5 cm (2) 8 cm
   (3) 4 cm (4) 5 cm

4. A projectile is projected from origin of a co-ordinate system with initial velocity of $(2\hat{i} + 3\hat{j})$m/s. The range of the projectile will be (g = 10 m/s^2($-\hat{j}$))
   (1) 4 m (2) 1.2 m
   (3) 2.4 m (4) 3.6 m

5. Which of the following statement is incorrect regarding the motion in a plane?
   (1) A body can't move on a curved path with constant acceleration
   (2) A body can move on a curved path with constant acceleration
   (3) The angle between acceleration and velocity can be 90° or less than 90°
   (4) The angle between acceleration and velocity can be other than 90°

6. **Statement A:** Two particles are projected with same speed but at angles of projection $(45° - \theta)$ and $(45° + \theta)$. Then their horizontal range will be same.
   Statement B: In oblique projectile, at the point of projection the angle between acceleration and velocity is acute.
   (1) Both statement (A) and statement (B) are correct
   (2) Both statement (A) and statement (B) are incorrect
   (3) statement (A) is correct and statement (B) is incorrect
   (4) statement (A) is incorrect and statement (B) is correct

7. A body is undergoing uniform circular motion then which of the following quantity is constant.
   (1) Linear velocity (2) Acceleration
   (3) Force (4) Angular velocity

8. If $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{B} = 4\hat{i} - \hat{k}$, then the unit vector in the direction of $\vec{A} - 2\vec{B}$ is

(1) $\frac{-2\hat{i}+3\hat{j}+7\hat{k}}{\sqrt{62}}$

(2) $\frac{\left(-6\hat{i}+3\hat{j}+8\hat{k}\right)}{\sqrt{109}}$

(3) $\frac{6\hat{i}+3\hat{j}+5\hat{k}}{\sqrt{70}}$

(4) $\frac{6\hat{i}-3\hat{j}+7\hat{k}}{\sqrt{94}}$

9. Starting from rest, a body slides down a 45° inclined plane in twice the time it takes to slide the same inclined plane in the absence of friction. What is the coefficient of friction between the body and the inclined plane?

(1) $\frac{1}{2}$ (2) $\frac{3}{4}$

(3) $\frac{3}{5}$ (4) $\frac{2}{3}$

10. A boy riding a bicycle with a speed of 4 m/s towards east direction, observes rain falling vertically downward. If he increases his speed to 8 m/s, rain appears to meet him at 45° to the vertical. The speed of rain with respect to ground is

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(1) 8 m/s (2) 4 m/s

(3) $4\sqrt{2}$ m/s (4) 2 m/s

11. The resultant of two forces cannot be more than the

(1) Average of the magnitude of forces

(2) Sum of magnitude of two forces

(3) Difference of magnitude of two forces

(4) Magnitude of any single force

12. A body of weight 20 N is on a horizontal surface, minimum force required to pull it when force makes an angle 60° with horizontal (angle of friction α = 30°) is

(1) 20 N (2) 10 N

(3) $\frac{20}{\sqrt{3}}$ N (4) $20\sqrt{3}$N

13. Determine the maximum acceleration of the train in which a box lying on its floor will remain stationary. Given the coefficient of friction between box and train's floor is 0.15

(1) 12 m s^{-2} (2) 1.5 m s^{-2}

(3) 1.7 m s^{-2} (4) 0.7 m s^{-2}

14. Which among the following statement is **incorrect**?

(1) Tension in the string is always away from body on which it acts

(2) Tension in the string can be more than the weight attached at the end of the string

(3) Tension in the string is always same throughout the length of the string

(4) Tension can only pull an object but cannot push

15. A particle of mass M is revolving along a circle of radius R and another particle of mass M is revolving in a circle of radius r. If time period of both particles are same, then the ratio of their angular velocities is

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(1) 1 (2) $\frac{r^2}{R^2}$

(3) $\frac{R}{r}$ (4) $\sqrt{\frac{R}{r}}$

16. A particle A is projected with speed V_A from a point making an angle 60° with the horizontal. At the same instant, a second particle B is thrown vertically upwards from a point directly below the maximum height point of parabolic path of A, with velocity V_B. If the two particles collide then the ratio of V_A / V_B should be

(1) 1 (2) $\sqrt{3}$

(3) $\frac{2}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2}$

17. A car travels 40 km due east after then 30 km due north. Displacement of the car will be

(1) 70 km, north east (2) 50 km, 37° north of east

(3) 50 km, 53° north of east (4) 70 km, 37° north of east

18. A particle starts moving with constant acceleration and with initial velocity of $\left(2\hat{i}+2\hat{j}\right)$ m/s. After 2 s from start its velocity becomes $\left(4\hat{i}+6\hat{j}\right)$ m/s. The magnitude of its acceleration is

(1) $\sqrt{6}$ m/s^2 (2) $\sqrt{3}$ m/s^2

(3) $\sqrt{5}$ m/s^2 (4) 2 m/s^2

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19. The vector $\vec{P}$ makes an angle of 120° with the x-axis and the vector $\vec{Q}$ makes angle of 30° with the y-axis. Their resultant will be

(1) $P + Q$ (2) $\sqrt{P^2 - Q^2}$

(3) $\sqrt{P^2 + Q^2}$ (4) $P + \sqrt{3}Q$

20. A person is swimming towards south at 10 km/h and the river current is flowing at 6 km/h in the direction of 60° east of north. Then the velocity of person with respect to the ground and the angle made with south direction, in which person going is

(1) $\sqrt{50}$ km / h and $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(2) $\sqrt{76}$ km / h and $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{7}\right)$

(3) $\sqrt{76}$ km / h and $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(4) $\sqrt{50}$ km / h and $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{7}\right)$

21. A turn of radius 500 m is banked for vehicles going with optimum speed of 180 km/h. What is banking angle of this path?

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(1) $\tan^{-1}(2)$ (2) $\tan^{-1}(1)$

(3) $\tan^{-1}(1/2)$ (4) $\tan^{-1}(0.33)$

22. A ball of mass 2 kg strikes a wall with speed of 10 m/s at an angle 30° to the surface of the wall. Ball rebounds with same speed at the same angle. The magnitude of change in the momentum of the ball is

(1) 20 N s (2) 10 N s

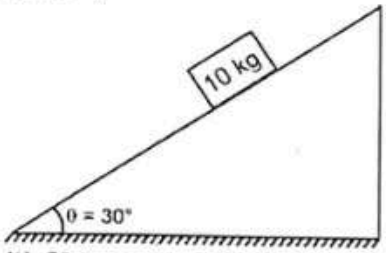
(3) Zero (4) $20\sqrt{3}$ N s

23. A monkey is descending from the branch of a tree with a constant acceleration. If the breaking strength of the branch is 50% of the weight of the monkey, then minimum acceleration with which the monkey can slide down without breaking the branch is

(1) g (2) $\frac{g}{2}$

(3) $\frac{g}{4}$ (4) $\frac{3g}{4}$

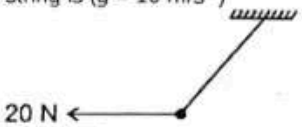
24. A block of mass 10 kg is kept on a fixed rough inclined plane having $\mu = 0.8$. The frictional force acting on the block is (g = 10 ms^{-2})



(1) 50 N (2) 60 N

(3) $40\sqrt{3}N$ (4) $50\sqrt{3}N$

25. A particle of mass m = 1 kg is tied to a string and a horizontal force F = 20 N is applied on the particle as shown in figure. If the particle is in equilibrium, the tension in the string is (g = 10 m/s^2)



(1) $5\sqrt{10}N$ (2) $5\sqrt{30}$N

(3) $3\sqrt{20}$N (4) $10\sqrt{5}$N

26. **Assertion (A)** : A graph is plotted between magnitude of force acting on a particle (taken on Y-axis) and time (taken on X-axis). Area under the graph represents impulse.
Reason (R) : Impulse equals magnitude of change in momentum.

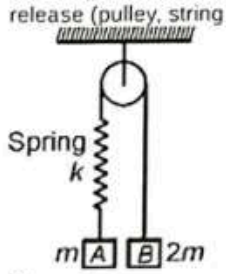
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

27. A car M right now on the cross-road is moving towards west at 40 km/h. Another car P which is right now at 5 km south of the crossing is moving towards the crossing at 40 km/h. The closest distance of approach between the two cars will be

(1) 5 km (2) 2.5 km

(3) $\frac{5}{\sqrt{2}}$km (4) $5\sqrt{2}$km

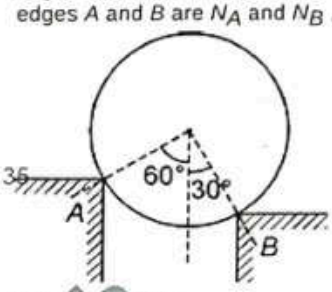
28. Two blocks A and B of masses m and $2m$ respectively are held at rest such that the spring is in natural length. Find out the accelerations of blocks A and B respectively just after release (pulley, string and spring are massless).



(1) g ↓, g ↓ (2) 0, 0

(3) g ↓, 0 (4) $\frac{g}{3}$ ↓, $\frac{g}{3}$ ↑

29. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in figure. If normal reaction at the edges A and B are N_A and N_B, then (Neglect friction)



(1) $N_A = \sqrt{2}N_B$ (2) $N_A = \frac{Mg}{2}$

(3) $N_B = \sqrt{3}N_A$ (4) Both (2) and (3)

30. Match entries in Column-I with entries in Column-II and choose the correct option.

Column-I		Column-II	
A.	Limiting friction	P.	Has magnitude $\mu_k N$
B.	Static friction	Q.	Maximum value of static friction
C.	Kinetic friction	R.	Is a self adjusting force

(1) A → P, B → Q, C → R (2) A → R, B → P, C → Q

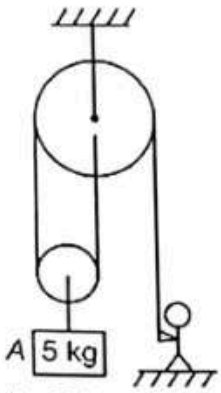
(3) A → Q, B → R, C → P (4) A → Q, B → P, C → R

31. Consider the following statements:
Statement (A): A body with constant magnitude of acceleration may not speed up.
Statement (B): A body with constant acceleration speeds up at every instant of its motion.
The correct option is

(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both statements are correct
(4) Both statements are incorrect

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32. To keep the block A of mass 5 kg in equilibrium, the man pulls the rope by a force F as shown in the figure. The magnitude of force F is (g = 10 m/s^2)



(1) 25 N (2) 50 N
(3) 100 N (4) 15 N

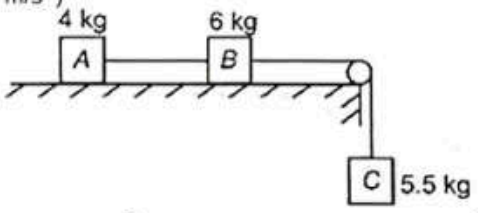
33. A body moves along a circular path of radius 5 m and the coefficient of friction is 0.5. The maximum magnitude of velocity so that body will not skid is
(1) 4 m/s (2) 5 m/s
(3) 6 m/s (4) 8 m/s

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34. If the resultant of two vectors, where one vector is double in magnitude than the other, is perpendicular to vector of smaller magnitude, then the angle between the vectors is
(1) 160° (2) 135°
(3) 120° (4) 150°

35. Two balls are moving on the same smooth horizontal plane. Their velocity components along one edge of the square plane are 10 $\sqrt{3}$& 20 m/s. Their velocity components along the perpendicular edge are 30 m/s & 20 m/s respectively. Find the angle between their directions of motion.
(1) 15° (2) 45°
(3) 30° (4) 60°

36. Three blocks A, B and C are placed on a rough horizontal surface. Friction coefficient between blocks and surface is 0.6. Acceleration of the block C in given situation is (g = 10 m/s^2)



(1) 3.5 m/s^2 (2) 0.5 m/s^2
(3) 4.5 m/s^2 (4) Zero

CC-135

37. A gun throws a shell with speed of 98 m/s when the elevation is 45°, the range is found to be 950 m. How much range is decreased by air resistance? (g = 9.8 m/s^2)
(1) 20 m (2) 30 m
(3) 40 m (4) 50 m

38. The given vector $\frac{3\hat{i}}{5}+\frac{4\hat{j}}{5}+b\hat{k}$ is unit vector. What is the value of b?
(1) Zero (2) 1
(3) $\frac{4}{3}$ (4) $\frac{1}{3}$

39. A projectile is fired with velocity u making angle θ with the horizontal. What is the magnitude of change in velocity when it is at level of projection?
(1) $u\cos\theta$ (2) u
(3) $2u\sin\theta$ (4) $u\cos\theta - u$

40. Momentum of a block as a function of time is given by $\vec{p}$= ($10t^2\hat{i} + 5t\hat{j}$) kg ms^{-1}. The force acting on block at t = 2 s is
(1) $\vec{F}$= (10$\hat{i}$ + 5$\hat{j}$) N (2) $\vec{F}$= (20$\hat{i}$ + 6$\hat{j}$) N
(3) $\vec{F}$= (40$\hat{i}$ + 5$\hat{j}$) N (4) $\vec{F}$= (5$\hat{i}$ + 10$\hat{j}$) N

41. Two particles A and B are projected with same speed so that ratio of maximum height reached is 3 : 1. If the speed of A is doubled without altering other parameter, the ratio of the horizontal range obtained by A and B
(1) 1 : 1 (2) 4 : 1
(3) 2 : 1 (4) 3 : 2

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42. Consider the following statements.
A. Frictional force always opposes motion.
B. Coefficient of static friction is greater than coefficient of kinetic friction for given pair of surface.
C. Frictional force is non conservative force.
Based on above information pick the correct statement.
(1) Only B is true (2) Only B and C are true
(3) Only C is true (4) All statements are true

43. The radii of circular paths of two particles of same mass are in ratio 3 : 4, then what will be ratio of speed if they have same magnitude of centripetal force?
(1) 2 : 3 (2) $\sqrt{3}$: 2
(3) $\sqrt{3}$: 1 (4) 4 : $\sqrt{3}$

44. A cannon ball has same horizontal range corresponding to maximum height of 16 m and 9 m reached by the projectile with same projection speed. The correct value of horizontal range is
(1) 24 m (2) 48 m
(3) 144 m (4) 192 m

45. A vehicle of mass 2000 kg starts moving on a circular track of radius 100 m with increasing speed at rate 4 m/s^2. The ratio of tangential acceleration to net acceleration at 20 second is
(1) 1 : 16 (2) 4 : $\sqrt{257}$
(3) 1 : $\sqrt{257}$ (4) $\sqrt{257}$: 4

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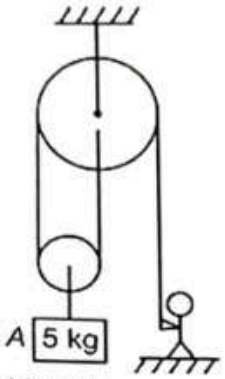
CHEMISTRY

46. The number of lines obtained in UV region of hydrogen spectrum when electron jumps from 6th excited state to 1st excited state is/are:
(1) 5 (2) Zero
(3) 4 (4) 3

47. Calculate the wave number (m^{-1}) if, the frequency of violet radiation is 7.5 × 10^{14} Hz
(1) 2.5 × 10^6 (2) 4 × 10^{-7}
(3) 1.33 × 10^{-15} (4) 2.5 × 10^7

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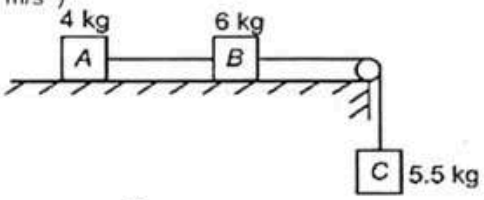
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CHEMISTRY

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48. Which experimental phenomenon is explained by wave nature of electromagnetic radiation?
(1) Diffraction
(2) Black-body radiation
(3) Line spectra of hydrogen atom
(4) Photoelectric effect

49. Consider the following statements about photoelectric effect
(a) There is no time lag between the striking of light beam and the ejection of electrons from the metal surface.
(b) The number of electrons ejected is proportional to the intensity of light.
(c) The kinetic energy of the ejected electrons increases with increase of frequency of the light used.
The correct statements are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

50. The correct order of wavelength of given electromagnetic waves is
(1) Microwave > UV > IR > Radio waves
(2) Radiowaves > Microwave > UV > IR
(3) Microwave > Radiowave > IR > UV
(4) Radiowaves > Microwave > IR > UV

51. If a 30 watt bulb emits monochromatic light of wavelength 1320 nm, then the number of photons emitted per second by the bulb will be (Given $h = 6.6 \times 10^{-34}$ Js)
(1) 4×10^{21}
(2) 2×10^{20}
(3) 2×10^{22}
(4) 4×10^{19}

52. The energies E_1 and E_2 of two monochromatic radiations are 5 eV and 10 eV respectively. The relation between their wavelengths λ_1 and λ_2 will be
(1) $\lambda_1 = 2\lambda_2$
(2) $\lambda_1 = \lambda_2$
(3) $\lambda_1 = \frac{\lambda_2}{2}$
(4) $4\lambda_1 = \lambda_2$

53. If the wavelength of the first line of He^+ ion in Lyman series is x Å, then the wavelength of the first line of Li^{2+} in Balmer series will be
(1) $\frac{2x}{9}$ Å
(2) $\frac{12x}{5}$ Å
(3) 3 x Å
(4) $\frac{x}{9}$ Å

54. Splitting of spectral lines under the influence of electric field and magnetic field are respectively called
(1) Zeeman effect and Stark effect
(2) Stark effect and photoelectric effect
(3) Stark effect and Zeeman effect
(4) Photoelectric effect and Zeeman effect

55. When electromagnetic radiation of wavelength 300 nm falls on the surface of a metal, electrons are emitted with the kinetic energy of 1.68×10^5 J mol^{-1}. What is the minimum energy needed to remove an electron from the metal?
($h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ ms^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1})
(1) 2.31×10^5 J mol^{-1}
(2) 2.31×10^6 J mol^{-1}
(3) 3.84×10^4 J mol^{-1}
(4) 3.84×10^{-19} J mol^{-1}

56. The number of protons, neutrons and electrons in $^{138}_{57}La$ respectively are
(1) 57, 82 and 57
(2) 57, 57 and 81
(3) 82, 57 and 57
(4) 57, 81 and 57

57. How many maximum electrons can be accommodated in an orbital for which n = 4 and l = 2?
(1) 5
(2) 10
(3) 2
(4) 6

58. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		Column II (orbital)
(a)	n = 2, l = 1	(i)	3*p*
(b)	n = 3, l = 2	(ii)	3*d*
(c)	n = 1, l = 0	(iii)	2*p*
(d)	n = 3, l = 1	(iv)	1*s*

(1) (a)(iii), (b)(i), (c)(iv), (d)(ii)
(2) (a)(iii), (b)(ii), (c)(iv), (d)(i)
(3) (a)(ii), (b)(i), (c)(iii), (d)(iv)
(4) (a)(i), (b)(ii), (c)(iii), (d)(iv)

59. If a particle of mass 20 mg is moving with a velocity of 500 ms^{-1} then the de Broglie wavelength associated with the moving particle is ($h = 6.6 \times 10^{-34}$ Js)
(1) 3.3×10^{-35} m
(2) 6.6×10^{-32} m
(3) 6.6×10^{-35} m
(4) 3.3×10^{-36} m

60. Orbital angular momentum value of an electron for *p* orbital is
(1) Zero
(2) $\sqrt{3}\frac{h}{2\pi}$
(3) $\frac{h}{2\sqrt{2}\pi}$
(4) $\sqrt{2}\frac{h}{2\pi}$

61. Which of the following sets of quantum numbers is not possible?
(1) n = 4, ℓ = 3, m = +3, s = $-\frac{1}{2}$
(2) n = 4, ℓ = 2, m = +3, s = $+\frac{1}{2}$
(3) n = 4, ℓ = 2, m = –2, s = $+\frac{1}{2}$
(4) n = 4, ℓ = 3, m = –2, s = $+\frac{1}{2}$

62. In the given configuration representation which rule is violated?



(A) Aufbau principle
(B) Pauli's exclusion principle
(C) Hund's rule of maximum multiplicity
Choose the correct option.
(1) (A), (B) and (C)
(2) (A) and (C) only
(3) (B) and (C) only
(4) (B) only

63. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.
(1) a(ii), b(iv), c(i), d(iii)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(iii), b(i), c(iv), d(ii)

64. A ball of mass 10 g is moving with a speed of 100 m/s. If the uncertainty in speed is 1%, then the uncertainty in the position of the ball is
(1) 5.25×10^{-29} m
(2) 5.25×10^{-33} m
(3) 5.25×10^{-34} m
(4) 5.25×10^{-31} m

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65. The maximum degenerate orbitals in 3rd shell of Li^{2+} is
(1) 1 (2) 3
(3) 7 (4) 9

66. Two electrons occupying the same orbital are distinguished by
(1) Principal quantum number
(2) Azimuthal quantum number
(3) Magnetic quantum number
(4) Spin quantum number

67. Match List-I with List-II.

List-I (Atomic number)	List-II (Period and group in periodic table)
(a) 49	(i) 6th period of 3rd group
(b) 92	(ii) 5th period and 13th group
(c) 38	(iii) 5th period and 2nd group
(d) 57	(iv) 7th period and 3rd group

Choose the correct option.
(1) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(4) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

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68. The correct order of first ionization enthalpy for the given elements is
(1) Be < B < C < N < O < F (2) B < Be < C < O < N < F
(3) Be < B < C < O < N < F (4) B < Be < C < N < O < F

69. The correct order of negative electron gain enthalpy is
(1) Se < S < Br < F (2) Se < F < S < Br
(3) F < Br < Se < S (4) F < Se < S < Br

70. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ respectively are
(1) +2 and 5 (2) +3 and 3
(3) +3 and 5 (4) +3 and 6

71. The pair of elements which have same value of electronegativity on Pauling scale, is
(1) Be and B (2) C and S
(3) Al and P (4) Na and Mg

72. Consider the following statements:
(a) Symbol of the element having atomic number 105 is Unp
(b) S^{2-}, K^+ and Cl^- are isoelectronic species
(c) The first ionisation enthalpy of element with atomic number 19 is higher than that of the element with atomic number 11
(d) IUPAC official name of atomic number 106 is seaborgium
The correct statements are
(1) (a), (b) and (c) only (2) (b), (c) and (d) only
(3) (a), (c) and (d) only (4) (a), (b) and (d) only

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73. Match column I with column II

Column-I (Compound)	Column-II (Nature)
a. Na_2O	(i) Neutral
b. Cl_2O_7	(ii) Acidic
c. Al_2O_3	(iii) Basic
d. CO	(iv) Amphoteric

The correct match is
(1) a(iii), b(ii), c(iv), d(i) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(iv), c(ii), d(i)

74. The correct order of ionic radii among the following species is
(1) $F^- < O^{2-} < Na^+ < Mg^{2+}$ (2) $O^{2-} < F^- < Na^+ < Mg^{2+}$
(3) $Mg^{2+} < Na^+ < F^- < O^{2-}$ (4) $Na^+ < Mg^{2+} < O^{2-} < F^-$

75. Which among the following processes is an endothermic process?
(1) $F(g) \rightarrow F^-(g)$ (2) $Br(g) \rightarrow Br^-(g)$
(3) $O^-(g) \rightarrow O^{2-}(g)$ (4) $O(g) \rightarrow O^-(g)$

76. The correct increasing order of atomic radii for the following species is
(1) Be < Li < Mg < Na (2) Li < Be < Mg < Na
(3) Li < Be < Na < Mg (4) Be < Li < Na < Mg

77. The correct order of metallic character is
(1) Na < Mg < Be < Si < P (2) Be < Mg < Na < P < Si
(3) P < Si < Be < Mg < Na (4) P < Be < Si < Mg < Na

78. Which pair represents metalloids?
(1) Tellurium and thorium (2) Ruthenium and iridium
(3) Arsenic and tellurium (4) Beryllium and aluminium

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79. The first ionization potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be
(1) –5.1 eV (2) –9.1 eV
(3) + 10.2 eV (4) –2.5 eV

80. If the kinetic energy of a particle is tripled, its de-Broglie's wavelength becomes
(1) 2 times (2) 4 times
(3) $\sqrt{2}$ times (4) $\frac{1}{\sqrt{3}}$ times

81. For multielectronic species, the correct order of energy of the given orbitals is
(1) 3d > 4p > 5s > 5p (2) 5p > 5s > 4p > 3d
(3) 4p > 3d > 5p > 5s (4) 5s > 4p > 3d > 5p

82. The total number of radial nodes in 4p orbital is
(1) 2 (2) 1
(3) Zero (4) 3

83. Find the formula of halide of a metal whose successive ionization enthalpies are x, 2x, 5x, 100x kJ mol^{-1} respectively
(1) MX (2) MX_2
(3) MX_3 (4) M_2X

84. Halogens are present in which group of periodic table?
(1) 14 (2) 15
(3) 16 (4) 17

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85. The kinetic energy of an electron in 3rd Bohr orbit of He^+ ion is
(1) 8.2 eV (2) 7.1 eV
(3) 9.3 eV (4) 6.04 eV

86. The Bohr orbit radius of hydrogen atom in the ground state is 0.53 Å. The radius of Be^{3+} ion in 1st excited state is
(1) 1.06 Å (2) 0.53 Å
(3) 1.59 Å (4) 2.12 Å

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87. Consider the following statements.
(a) According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.
(b) The de Broglie wavelength of an electron of H-atom in the 3rd Bohr orbit is $6\pi a_0$ (a_0 is the Bohr orbit)
(c) Angular momentum of an electron in the 4th orbit of hydrogen atom is $\frac{2h}{\pi}$.
The correct statement(s) is/are
(1) (c) only
(2) (b) and (c) only
(3) (a) and (b) only
(4) (a), (b) and (c)

88. According to the Bohr theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon?
(1) n = 5 to n = 1
(2) n = 4 to n = 2
(3) n = 4 to n = 3
(4) n = 5 to n = 3

89. Consider the following statements
(a) Gallium is also known as Eka-silicon
(b) Fourth period of modern periodic table contains 18 elements
(c) The elements with atomic number more than 92 are called transuranium elements
The correct statements are
(1) (b) and (c) only
(2) (a) and (b) only
(3) (a) and (c) only
(4) (a), (b) and (c)

90. Given below are two statements; one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A) : The second ionisation enthalpy of an element will be lesser than the first ionisation enthalpy.
Reason (R) : It is more easier to remove an electron from a positively charged ion than from the neutral atom.
In the light of the above statements, choose the correct answer.
(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)
(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

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91. Identify the attribute that makes systematics, a wider field of science than taxonomy.
(1) Categorisation of different organisms
(2) Identification, nomenclature and classification of organisms
(3) Dealing with diversity of organisms along with comparative studies
(4) Taking into account, the phylogenetic relationships.

92. ICBN assigns rules for the scientific naming of
(1) Bacteria
(2) Plants
(3) Viruses
(4) Insects

93. Which of the following is **not** true?
(1) Genera are aggregates of closely related species.
(2) Plant family Convolvulaceae is included in the order Polymoniales, mainly on the basis of vegetative characters.
(3) The organisms with dorsal hollow neural system belong to the phylum, Chordata.
(4) Lower the taxa, more are the characteristics that the members within the taxon share.

94. Match the following columns and select the **correct** option.

	Column I	Column II
a.	G_1 phase	(i) Cytoplasmic growth
b.	S phase	(ii) Interval between mitosis and initiation of DNA replication
c.	G_2 phase	(iii) Condensation of chromatin
d.	M phase	(iv) DNA replication

(1) a(iv); b(ii); c(iii); d(i)
(2) a(iv); b(ii); c(i); d(iii)
(3) a(ii); b(iv); c(i); d(iii)
(4) a(ii); b(iv); c(iii); d(i)

95. Mitosis occurs in all, **except**
(1) Somatic cells
(2) Cell lining of gut
(3) Meristematic tissue
(4) Pollen mother cell

96. Name the order in which felidae and canidae are placed.
(1) Carnivora
(2) Mammalia
(3) Primata
(4) Animalia

97. Book "*Systema Naturae*" was written by
(1) Aristotle
(2) Carolus Linnaeus
(3) R.H. Whittaker
(4) Carl Woese

98. Observe the following taxa associated with wheat:
a. Poales
b. Poaceae
c. Angiospermae
d. Plantae
e. Monocotyledonae
Which of the following options **correctly** depicts the sequential order of these taxa in ascending order?
(1) a → d → b → e → c
(2) b → a → e → c → d
(3) b → e → c → a → d
(4) c → a → d → b → e

99. Biodiversity is referred as
(1) Diverse environment of a particular species
(2) The number and types of organisms on earth
(3) Availability of different food resources for a specific organism
(4) Variation among the offspring of an organism

100. Karyokinesis is the division of
(1) Cytoplasm
(2) Nucleus
(3) Chloroplast
(4) Mitochondria

101. Classification is the process of
(1) Recognising the characteristic features of an organism
(2) Naming of living organisms
(3) Grouping of organisms into convenient categories based on some easily observable characters
(4) Discovering new species of plants and animals

102. How many obligate taxonomic categories is/are common between wheat and mango?
(1) 3
(2) 4
(3) 1
(4) 2

103. Biological concept of species was given by
(1) Ernst Mayr
(2) R. H. Whittaker
(3) Ernst Haeckel
(4) Carolus Linnaeus

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104. Some cells in the adult animals that do not appear to exhibit division, e.g., heart cells, exit G_1-phase to enter the stage called
(1) S-phase (2) M-Phase
(3) Quiescent stage (4) G_2-phase

105. Which of the following stages lasts for months or years in the oocytes of some vertebrates?
(1) Pachytene (2) Diplotene
(3) Diakinesis (4) Telophase I

106. The final stage of prophase-I is marked by
(1) Terminalisation of chiasmata
(2) Formation of synaptonemal complex
(3) Appearance of recombination nodule
(4) Dissolution of the synaptonemal complex

107. Metaphase I is different from metaphase II as in former
(1) Bivalents align at equator
(2) Each chromosome is attached to the spindle fibres from both the poles
(3) Single equatorial plate is formed
(4) Each chromosome has single chromatid

108. In animal cell, cytokinesis
(1) Occurs via cell plate formation
(2) Is achieved by formation of furrow
(3) Progresses from centre to periphery
(4) Occurs centrifugally

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109. A cell has 20 bivalents in its zygotene stage. It means
(1) The cell has total 60 chromosomes
(2) Each haploid daughter cell has 20 chromosomes
(3) The cell has total 40 chromatids
(4) The cell has total 20 chromosomes

110. Match the column I with column II.

Column I	Column II
a. Dicotyledonae	(i) Phylum
b. Arthropoda	(ii) Class
c. Diptera	(iii) Division
d. Angiospermae	(iv) Order

Select the **correct** option.
(1) a(ii), b(i), c(iv), d(iii) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iii), b(ii), c(iv), d(i) (4) a(i), b(iii), c(iv), d(ii)

111. Interphase lasts about _____ of the duration of cell cycle.
(1) Less than 5% (2) 10% – 20%
(3) More than 95% (4) Less than 10%

112. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A): It becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio.
Reason (R): Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

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113. The phenomenon **absent** in an animal cell during interkinesis phase is
(1) RNA synthesis (2) Replication of DNA
(3) Protein synthesis (4) Centriole duplication

114. Enzyme recombinase activity is seen in
(1) Pachytene (2) Zygotene
(3) Leptotene (4) Diplotene

115. Liquid endosperm in coconut exemplifies a structure that is formed when
(1) Karyokinesis is not followed by cytokinesis
(2) Cytokinesis is achieved by cell plate formation
(3) Karyokinesis is delayed
(4) Four haploid uninucleate progenies are formed

116. Which among the following is not a part of the mitotic apparatus?
(1) Centrosomes (2) Microtubules
(3) Centromere (4) Spindle fibre

117. The term 'cell cycle' refers to
(1) DNA replication and cell growth in a coordinated way
(2) The sequence of events by which a cell duplicates its constituents and eventually divides into two daughter cells
(3) Duplication of chromosomes and then distribution of replicated chromosomes into the daughter nuclei
(4) A complex series of events through which the organelles synthesize their components and eventually duplicate

118. At the time of _____, organelles like mitochondria and plastids get distributed between the two daughter cells.
Choose the option which **correctly** fills in the above blank.
(1) Quiescent phase (2) Interphase
(3) Cytoplasmic division (4) Karyokinesis

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119. Microtubules which radiate out of centrosome are called
(1) Kinetochore (2) Asters
(3) Interzonal fibres (4) Phragmoplast

120. Duplication of semi-autonomous organelles occur in
(1) Both G_1 and G_2 (2) Both G_2 and M-phase
(3) G_1 phase (4) G_2 phase

121. A typical eukaryotic cell cycle is illustrated by human cells in culture. These cells divide once in approximately every
(1) 90 minutes (2) 24 minutes
(3) 24 hours (4) 90 hours

122. The longest phase of meiotic cell division is
(1) Prophase I (2) Metaphase I
(3) Prophase II (4) Metaphase II

123. Chiasmata or X-shaped structures become visible during
(1) Diplotene stage (2) Zygotene stage
(3) Pachytene stage (4) Leptotene stage

124. The structure **not** found in dividing cells of a higher plants is
(1) Metaphasic plate (2) Spindle fibres
(3) Centrioles (4) Cell plate

125. The best stage to study the morphology of chromosomes is
(1) Metaphase (2) Anaphase
(3) Telophase (4) Prophase

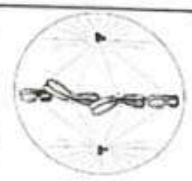
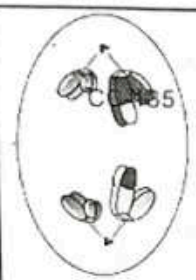
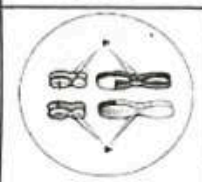
126. In animal cell, centriole duplication takes place during the phase in which
(1) Mitochondrial duplication takes place
(2) Actual cell division occurs
(3) Mitotic apparatus is formed
(4) DNA replication occurs

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127. In which of the following stages of meiosis-I, separation of homologous chromosomes occur?
(1) Prophase-I (2) Anaphase-I
(3) Metaphase-I (4) Telophase-I

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128. Match the stages of cell division given in **Column I** to their characteristic features given in **Column II** and select the **correct** option.

	Column I		Column II
(a)		(i)	Homologous chromosomes separate and move towards the opposite poles
(b)		(ii)	The bivalent chromosomes align on the equatorial plate
(c)		(iii)	Centromere splits and chromatids move towards opposite poles
(d)		(iv)	Chromosomes are moved to spindle equator and get aligned through spindle fibres

(1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

129. Read the following statements and select the 'Correct' option stating which ones are true (T) and which ones are false (F)
A) Short preparatory phase between Meiosis-I and Meiosis-II is known as interkinesis
B) Synapsis occurs during Zygotene stage
C) Chromosomes donot completely decondense in Telophase-I
D) Diplotene represents congression of bivalents to the equatorial plate

A B C D
(1) T T T F
(2) T T T T
(3) F T T F
(4) T F F T

(1) (1) (2) (2)
(3) (3) (4) (4)

130. Identify the **correct** sequence of the stages of cell cycle.
(1) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$ (2) $G_0 \rightarrow G_1 \rightarrow G_2 \rightarrow S$
(3) $M \rightarrow S \rightarrow G_1 \rightarrow G_2$ (4) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

131. If a meiocyte has 40 chromosomes in its G_1 phase, then what will be the number of chromosomes in G_2 phase?
(1) 20 (2) 80
(3) 40 (4) 10

132. In which of the following aspects, prophase of mitosis is similar to prophase I of meiosis?
(1) Pairing of homologous chromosomes
(2) Enzyme mediated recombination process
(3) Formation of chromosome tetrads
(4) Condensation of chromatin materials

133. Select the option which is common for *Solanum*, *Petunia* and *Datura*.
(1) They belong to the family Solanaceae
(2) They have same specific epithet
(3) They belong to the same class as that of wheat
(4) They belong to the division dicotyledonae

134. Which among the following options w.r.t. universal rule of nomenclature of organisms is **not** true?
(1) Biological names are Latinised or derived from Latin irrespective of their origin.
(2) The first word in the biological name represents the genus while the second component denotes the binomial epithet..
(3) When biological names are handwritten then they are underlined separately.
(4) The first component of the name starts with a capital letter while the second component starts with small letter.

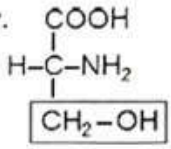
135. Read the following assertion (A) and reason (R) statements and select the **correct** option.
Assertion(A): The earliest classifications were based on the 'uses' of various organisms.
Reason(R): In early days, human beings needed to find sources for their basic needs of food, clothing and shelter from organisms.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

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136. How many nitrogen atom(s) is/are present in a lecithin?
(1) 4 (2) 3
(3) 2 (4) 1

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137. 

The amino acid represented by the structure given above is

(1) Glycine (2) Alanine
(3) Serine (4) Valine

138. Choose the **correct** option.
(1) Holoenzyme = Apoenzyme + coenzyme
(2) Holoenzyme = Apoenzyme – coenzyme
(3) Coenzyme = Apoenzyme + holoenzyme
(4) Coenzyme = Apoenzyme – holoenzyme

139. Which option best describes the tertiary level of protein structure?
(1) The association of multiple polypeptide chains through weak interactions.
(2) The specific three-dimensional conformation of a fully folded polypeptide, as exhibited by enzymes.
(3) The arrangement of a polypeptide chain into left-handed helices, just like an alpha helix or beta-pleated sheet.
(4) The overall shape formed when two or more polypeptide subunits come together, as seen in haemoglobin.

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140. Read the given statements and select the correct option.
Statement A: The type of co-factor found in peroxidase is actually an organic compound which is tightly bound to the apoenzyme.
Statement B: Catalase and carboxypeptidase possess same type of co-factor.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

141. Select the odd one w.r.t. aromatic amino acids.
(1) Tyrosine (2) Tryptophan
(3) Lysine (4) Phenylalanine

142. According to the general rule of thumb, if the rate of reaction becomes one-fourth of the initial, then what would be the new temperature of the reaction, if initially it was T?
(1) T + 20°C (2) T – 20°C
(3) T – 10°C (4) T + 10°C

143. Match column I with column II and choose the **correct** option.

Column I	Column II
a. Peptide bonds	(i) Cellulose
b. Nucleotide	(ii) Haemoglobin
c. Nucleoside	(iii) Adenylic acid
d. Glycosidic bonds	(iv) Uridine

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(i), c(iv), d(iii)
(3) a(i), b(ii), c(iv), d(iii) (4) a(ii), b(iii), c(iv), d(i)

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144. A double stranded DNA contains 20% of nitrogenous bases as guanine. What will be the expected percentage of other three nitrogenous bases i.e. cytosine, adenine and thymine represented as C, A and T?
(1) C - 20%, A - 30%, T - 30%
(2) C - 30%, A - 20%, T - 30%
(3) A - 40%, T - 20%, C - 20%
(4) A - 20%, T - 20%, C - 40%

145. Match column I with column II w.r.t. given compounds.

	Column I		Column II
a.	Adenine	(i)	1 heterocyclic ring
b.	Uracil	(ii)	4 homocyclic rings
c.	Cholesterol	(iii)	2 heterocyclic rings
d.	Adenylic acid	(iv)	3 heterocyclic rings

Select the **correct** option.
(1) a(iii), b(i), c(ii), d(iv) (2) a(i), b(iv), c(iii), d(ii)
(3) a(iv), b(iii), c(i), d(ii) (4) a(ii), b(iii), c(iv), d(i)

146. If we plot a graph for analysing the variation of change in temperature (x-axis) with enzyme activity (y-axis), the curve obtained will be
(1) Bell-shaped (2) Sigmoid
(3) Parabolic (4) Linear

147. 'X' is the prosthetic group of an enzyme named catalase. 'X' is also the prosthetic group in
(1) Carboxypeptidase (2) Peroxidase
(3) Carbonic anhydrase (4) Succinic dehydrogenase

148. Which of the following is least likely to be involved in stabilizing the three-dimensional tertiary structure of most proteins?
(1) Hydrogen bonds (2) Electrostatic interactions
(3) Hydrophobic interactions (4) Glycosidic bonds

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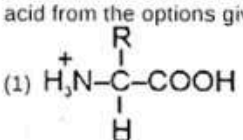
149. The most abundant protein in whole of the biosphere is
(1) Collagen (2) RuBisCO
(3) Chitin (4) Glycogen

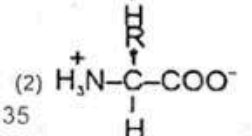
150. In a nucleic acid, the bond between the phosphate and hydroxyl group of a sugar is
(1) Peptide bond (2) Ester bond
(3) Glycosidic bond (4) Hydrogen bond

151. The enzyme which catalyses breaking of bonds between the glycerol and fatty acids in a molecule of triglyceride has enzyme commission number that starts with
(1) One (2) Two
(3) Three (4) Four

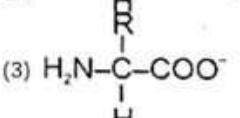
152. "All enzymes are proteins". An exception to this is
(1) Ribozyme (2) Dehydrogenase
(3) Hydrogenase (4) Lipase

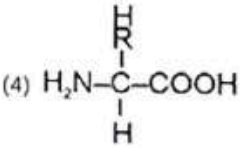
153. Select the **correct** structure of zwitterionic form of amino acid from the options given below.

(1) 

(2) 

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(3) 

(4) 

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154. Choose the **correct** statement w.r.t a competitive inhibitor.
(1) It binds irreversibly to the enzyme.
(2) Its effect can be overcome by decreasing the concentration of the enzyme.
(3) It resembles the substrate and competes with it for the active site on enzyme.
(4) It increases V_{max} of the enzyme-catalyzed reaction.

155. **Assertion(A):** The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.
Reason(R): The thoracic chamber is formed ventrally by the vertebral column, dorsally by the sternum, laterally by the ribs and on the lower side by the dome-shaped diaphragm.
In the light of above statements, select the **correct** option.
(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

156. Function not performed by conducting part of respiratory tract is
(1) Humidification of inhaled air
(2) Trapping of foreign particles
(3) Diffusion of O_2 and CO_2 between blood and tissues
(4) Bringing the inhaled air to the body temperature

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157. Complete the analogy by selecting the **correct** option w.r.t. respiratory structures.
Moist cuticle : Earthworm : : Lung : _____
(1) Aquatic insects (2) Aquatic mammals
(3) Fishes (4) Aquatic molluscs

158. Animals which have a network of tracheal tubes to transport atmospheric air within the body are
(1) Flatworms (2) Insects
(3) Coelenterates (4) Sponges

159. Choose the **odd** one w.r.t. forceful exhalation.
(1) Internal intercostals muscles contract
(2) Abdominal muscles contract
(3) Volume of thoracic cavity decreases
(4) External intercostal muscles contract

160. During swallowing, entry of food into larynx is prevented by a thin elastic cartilaginous flap called
(1) Glottis (2) Gullet
(3) Epiglottis (4) Uvula

161. Both human lungs are covered by_____ layered pleura.
Select the option that **correctly** fills the blank.
(1) Single (2) Double
(3) Triple (4) Tetra

162. During normal quiet inspiration
(1) External intercostal muscles relax
(2) Internal intercostal muscles contract
(3) Intra-pulmonary pressure is more than atmospheric pressure
(4) Volume of thoracic chamber increases in both antero-posterior and dorso-ventral axis

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163. Lungs have a large number of alveoli primarily for
(1) Having spongy texture and proper shape
(2) More surface area for diffusion of gases
(3) More space for increasing volume of inspired air
(4) More nerve supply

164. Structures which are **not** supported by incomplete cartilaginous rings are
(1) Tracheae (2) Primary bronchi
(3) Initial bronchi (4) Terminal bronchioles

165. Ribs and sternum are lifted upwards due to contraction of
(1) Diaphragm
(2) External inter-costal muscles
(3) Abdominal muscles
(4) Internal inter-costal muscles

166. The cartilaginous part/organ in humans which produces sound is
(1) Pharynx (2) Larynx
(3) Trachea (4) Uvula

167. The common passage for food and air is
(1) Pharynx (2) Bronchi
(3) Trachea (4) Eustachian tube

168. Breathing is exchange of gases between
(1) Atmospheric air and blood
(2) Atmospheric air and lungs
(3) Lungs and blood
(4) Tissue and blood

169. In normal respiration, contraction of which of the following muscles leads to a decrease in intra pulmonary pressure?
(1) Diaphragm
(2) Internal intercostal muscles
(3) Abdominal muscles
(4) Cardiac muscles

170. Which of the following pulmonary capacities/volume **cannot** be measured by spirometer?
(i) FRC
(ii) RV
(iii) TLC
Choose the correct option.
(1) Only (i) (2) (i), (ii) & (iii)
(3) Only (ii) & (iii) (4) Only (i) & (iii)

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171. Match column I with column II and choose the **correct** option

	Column I		Column II
a.	TV	(i)	1100 mL - 1200 mL
b.	IRV	(ii)	1000 mL - 1100 mL
c.	ERV	(iii)	500 mL
d.	RV	(iv)	2500 mL - 3000 mL

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(ii), b(i), c(iii), d(iv) (4) a(iii), b(iv), c(ii), d(i)

172. The volume of air which remains in lungs even after a forcible expiration is
(1) Residual volume
(2) Tidal volume
(3) Inspiratory reserve volume
(4) Expiratory reserve volume

173. If a healthy adult man inspires or expires 9000 mL of air per minute, then what will be his breathing rate per minute assuming his tidal volume remains normal?
(1) 14 times (2) 16 times
(3) 18 times (4) 19 times

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174. Total lung capacity of a normal human is represented by equation TLC = X + RV. X can be any of the lung volumes/capacities given below, **except**
(1) TV + FRC (2) TV + IRV + ERV
(3) IC + ERV (4) VC

175. _____ is the simplest amino acid.
Fill in the blank with a suitable option.
(1) Glycine (2) Alanine
(3) Valine (4) Isoleucine

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176. Triglycerides are made up of
(1) 2 fatty acids + 2 glycerol molecules
(2) 3 fatty acids + 2 glycerol molecules
(3) 3 fatty acids + 1 glycerol molecule
(4) 1 fatty acid + 3 glycerol molecules

177. Gingelly oil
(1) Is not an example of simple lipid
(2) Remains solid at room temperature
(3) Has lower melting point than ghee
(4) Considered strictly a biomacromolecule

178. Glottis is an opening of
(1) Pharynx
(2) Larynx
(3) Epiglottis
(4) Thyroid

179. Adult human haemoglobin consists of subunits of
(1) Two α and two β type
(2) One α and one β type
(3) Two α and one β type
(4) Three α and two β type

180. Cytidine is a/an
(1) Nucleoside
(2) Nucleotide
(3) Essential amino acid
(4) Non-essential amino acid

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